

NER (Apr 12, 2026)

Named Entity Recognition

Common Noun vs Proper Noun

✓ Capitalized

"Apple"

IOB
1 1
In Out
Beginning

There are negotiations
going on between US
and Iran in Islamabad
over the Strait of Hormuz.

OOOOOOB O B O B O O B I I

raw sentence → annotation
(OOOOOOB O B O B O O B I I)

Person — "Sushil Kumar"
Organization — "NITI Aayog"
Location — "Beijing", "Chennai",
"New York"

Time

→ "10:00 AM IST Apr 12, 2026"

Currency/Money — "\$5" "₹10 Million"

"₹2 Lakh Crore"

₹2,00,000 Crore

←

0 0 0 0 0 0 B-L 0 B-L 0 B-L

←

NEP as (promulgated) by VAC

was not acceptable for Tamil Nadu

(B-O 0 0 B-O 0 0 0 0)
(B-L I I I I)

←

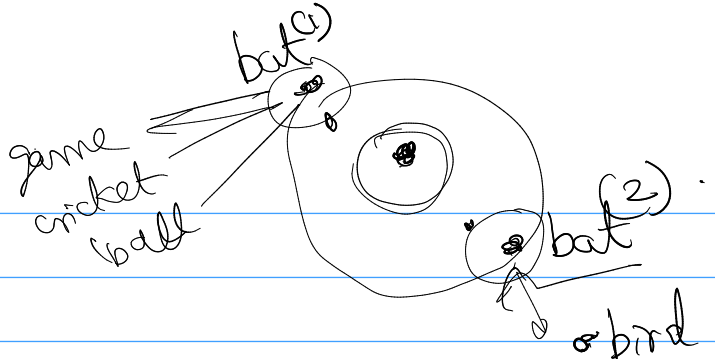
BERT

context-sensitive embeddings

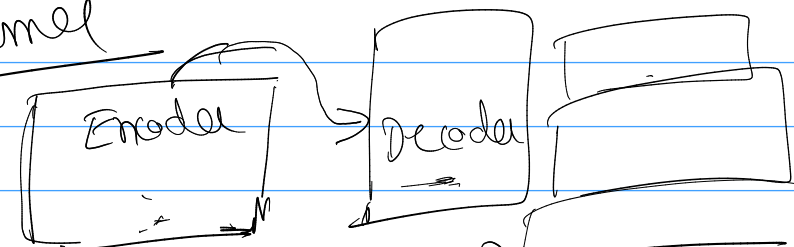
Glove
Word2Vec
→ fixed word
embeddings.

Kohli hit the ball hard with
the bat.

There was a swarm of bats
flying out when I went into
the cave.



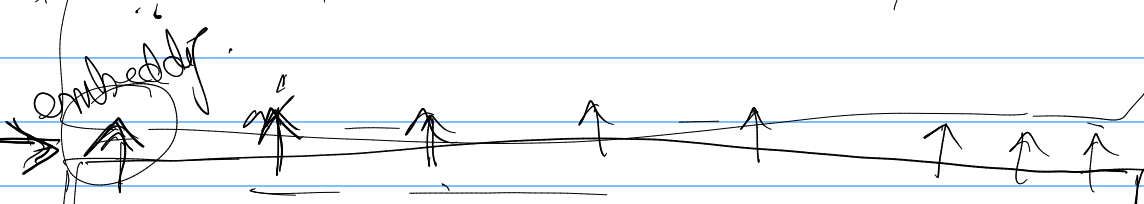
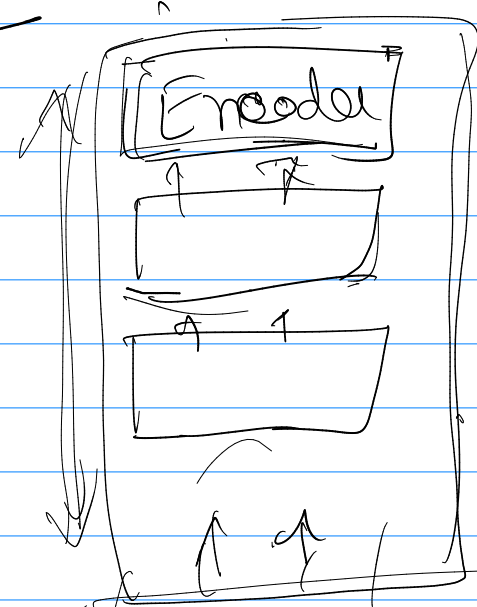
Transformer



cross attention

self attention

But

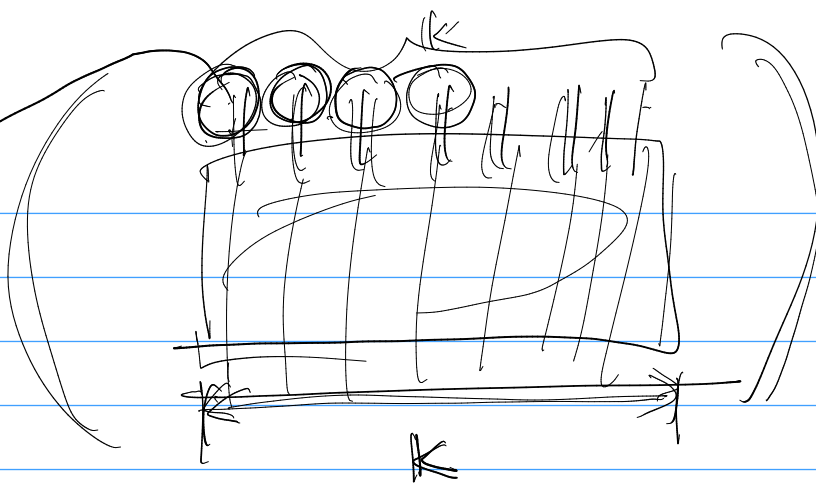


The batman hit the ball for a six

e_1 e_2 e_3 e_4 e_5 e_6 e_7 e_8

$0.6e_1 + 0.05e_2 + 0.05e_3 + \dots$

$0.6e_1 + 0.05e_2 + 0.05e_3 + \dots$



[CLS] [SEP]

tokens

CoNLL-2003 (hugface)
 NIST-IEER (nltk)

① Align "tokens" with word annotations.

Newcastle

(B=L) location.

(t1)

t2

(B-L)

I-~~I~~

New
 B-L

castle
 I

5+2

7

embed

classifier

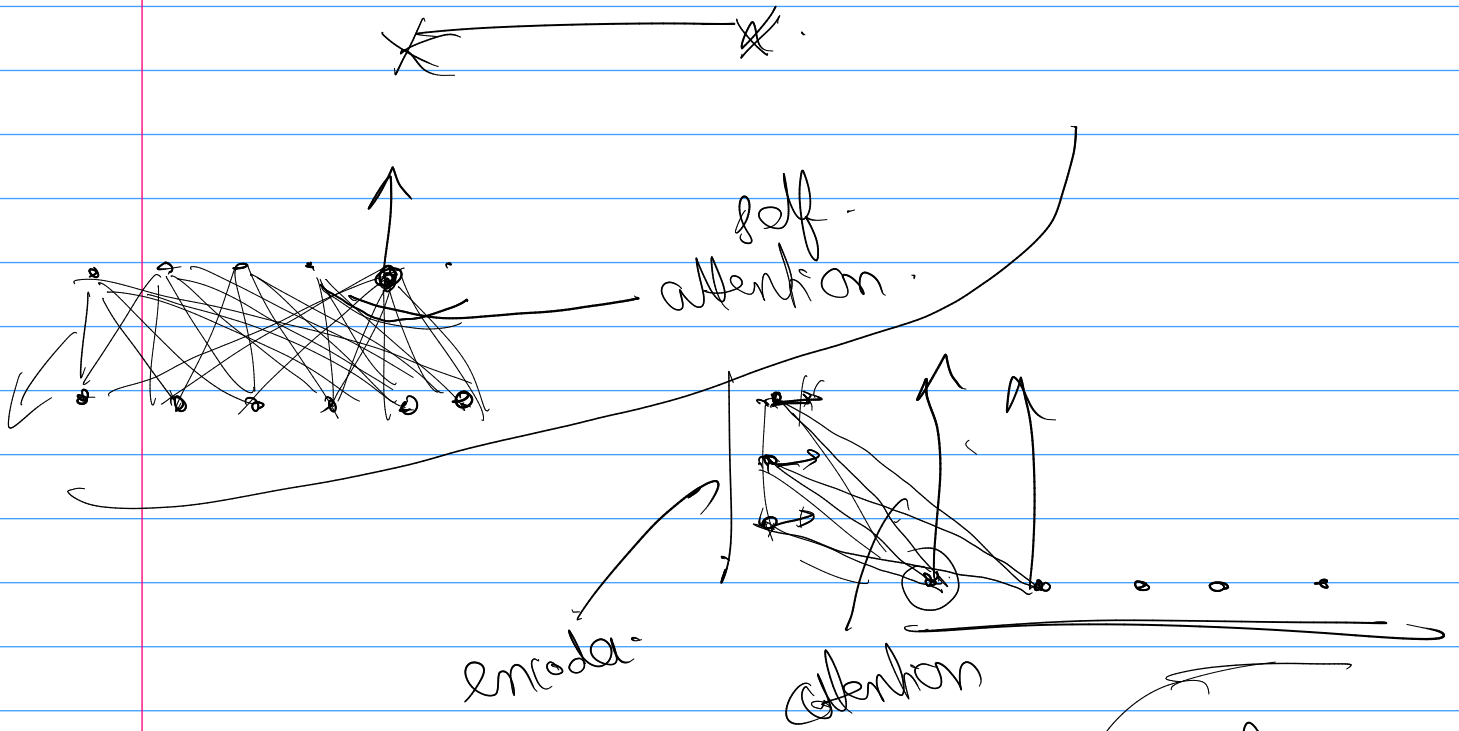
0.9

softmax

B-L

0
 0
 6
 0

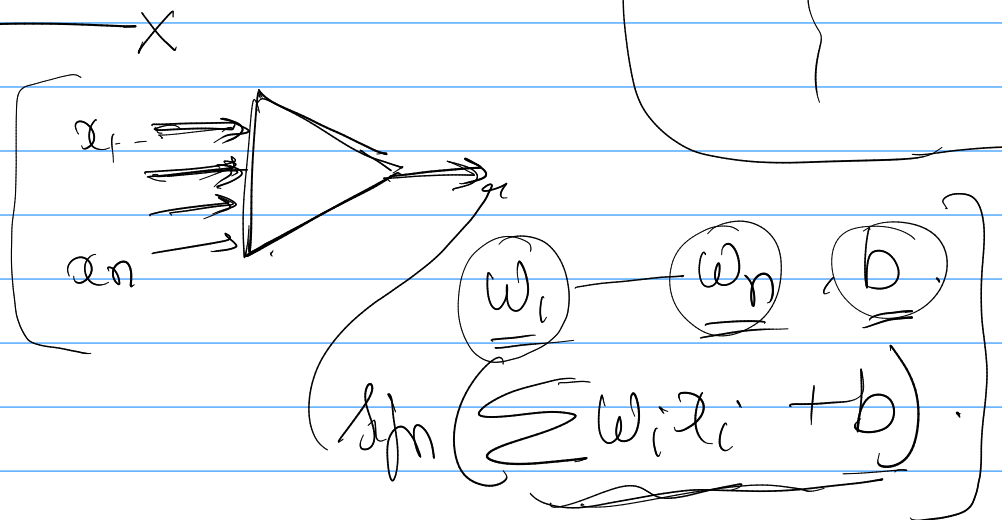
Loss: Cross-entropy.

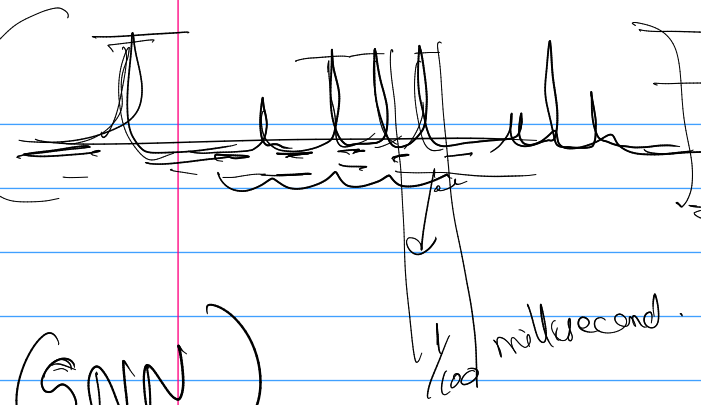
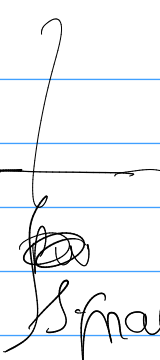


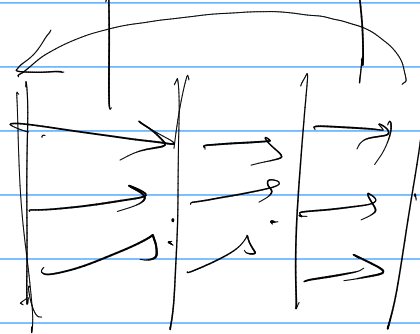
DL vs 'classical' ML
(Statistical).

SVM
Logistic Regression
Naive Bayes
Decision Trees

Perceptron



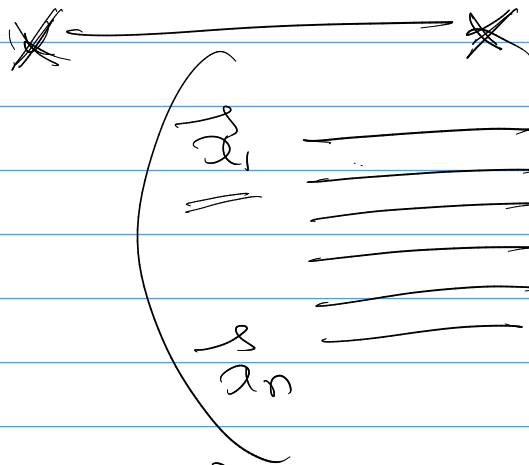
	Brain	ANN	
 (SNN) ↓ <u>Spiking</u> <u>Hodgkin-Huxley</u> <u>Connectionist Philosophy</u>	Pulses	Continuous signals	 Signal Elementary description size Architecture Structure
	Analog. Pulses	Digital / Binary	
	PDE's	Elementary	
	Much bigger	relatively small	
	Dynamic	Rigid	
<u>Plasticity</u> (connections $\times$ $\checkmark$) new	Chatter, Feedback loops	Casualty linear	



Neuro-morphic Computer ✓

ANNs

ML



$f^*(\vec{x}_i) = y_i$

$f(\vec{x})$: $\forall i: f(\vec{x}_i) \approx y_i$

Loss Fn: $L(f(\vec{x}), y)$

prediction ↓ Ground truth

ML problem

find f that minimizes

$$\sum_{i=1}^n L(f(\vec{x}_i), y_i)$$

among all $f \in S$

ac

No Free Lunch Theorem

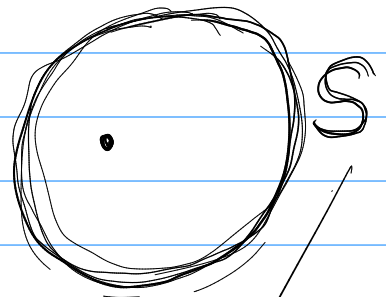
Exponential Quadratic Linear.

form-

$$\sum C_i z_i + b$$

$$\frac{1}{1 + e^{-(\sum C_i z_i + b)}}$$

→ linear



Function Family

DL	classical ML
no feature engineering.	(Feature Engineering.)
Depth	Shallow

Linear

$$\left(\sum C_i x_i + b \right)$$

function family.

parameters

$$5x_1 + 3x_2 + 9.6$$

$$10x_1 + 25.39x_2 + 9x_3 + 10.$$

Logic

$$\left(\frac{1}{1 + e^{-\left(\sum C_i x_i + b \right)}} \right)$$

(family).

$$F_{\vec{\theta}}$$

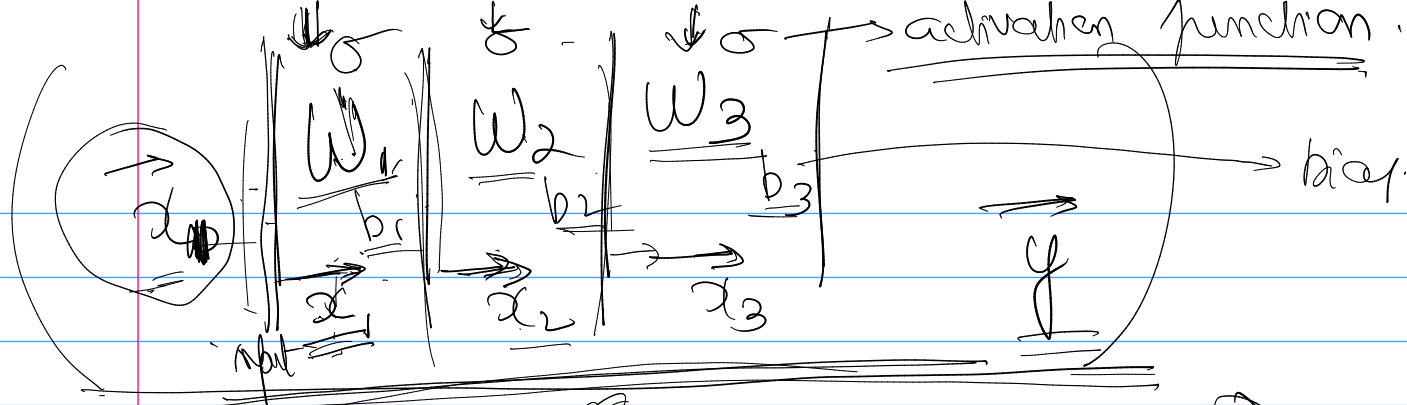
parameterization

Find $\vec{\theta}^*$ such that

$$\vec{\theta}^* = \arg \min_{\vec{\theta}} \sum_{i=1}^n L(f(\vec{x}_i; \vec{\theta}), y_i)$$

$$\left(f(\vec{x}^*; \vec{\theta}^*) \in \underline{F_{\vec{\theta}^*}} \right)$$

$x \longleftarrow ax$



$$y = \sigma(w_3 \cdot \sigma(w_2 \cdot \sigma(w_1 x_1 + b_1) + b_2) + b_3)$$

weight matrix

parameters: $(w_1, w_2, w_3, b_1, b_2, b_3)$

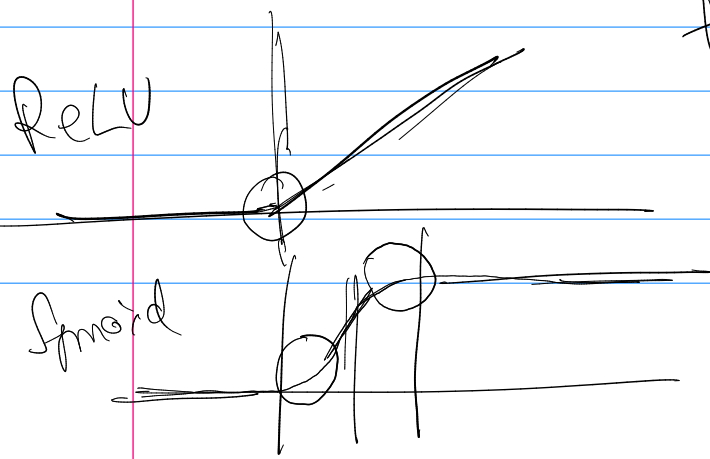
ReLU
sigmoid
tanh

Universal Approximation Theorem

Any "sufficiently complex NN" can

(≥ 2 layers) - enough neurons
with non-linear activation in each.

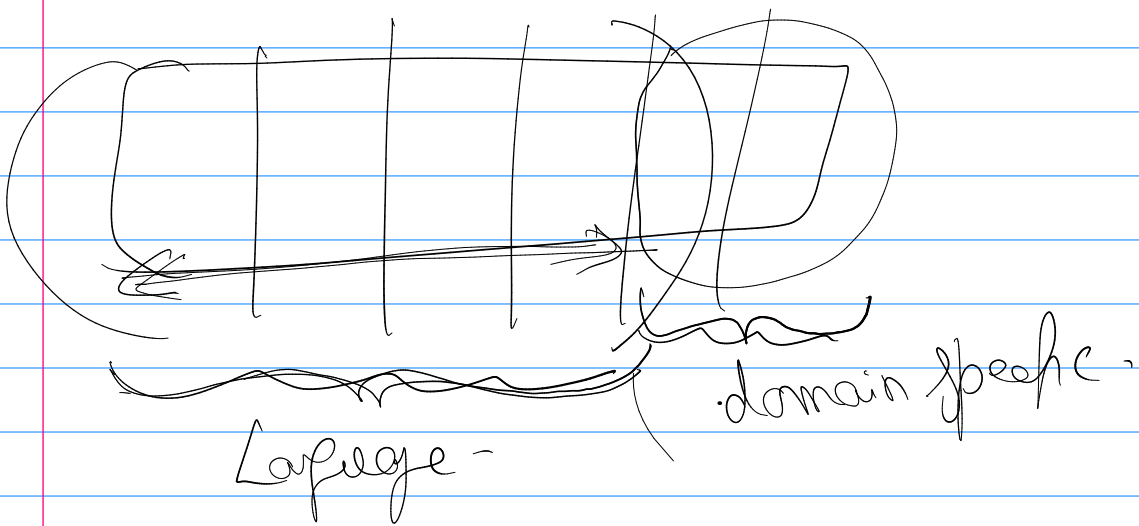
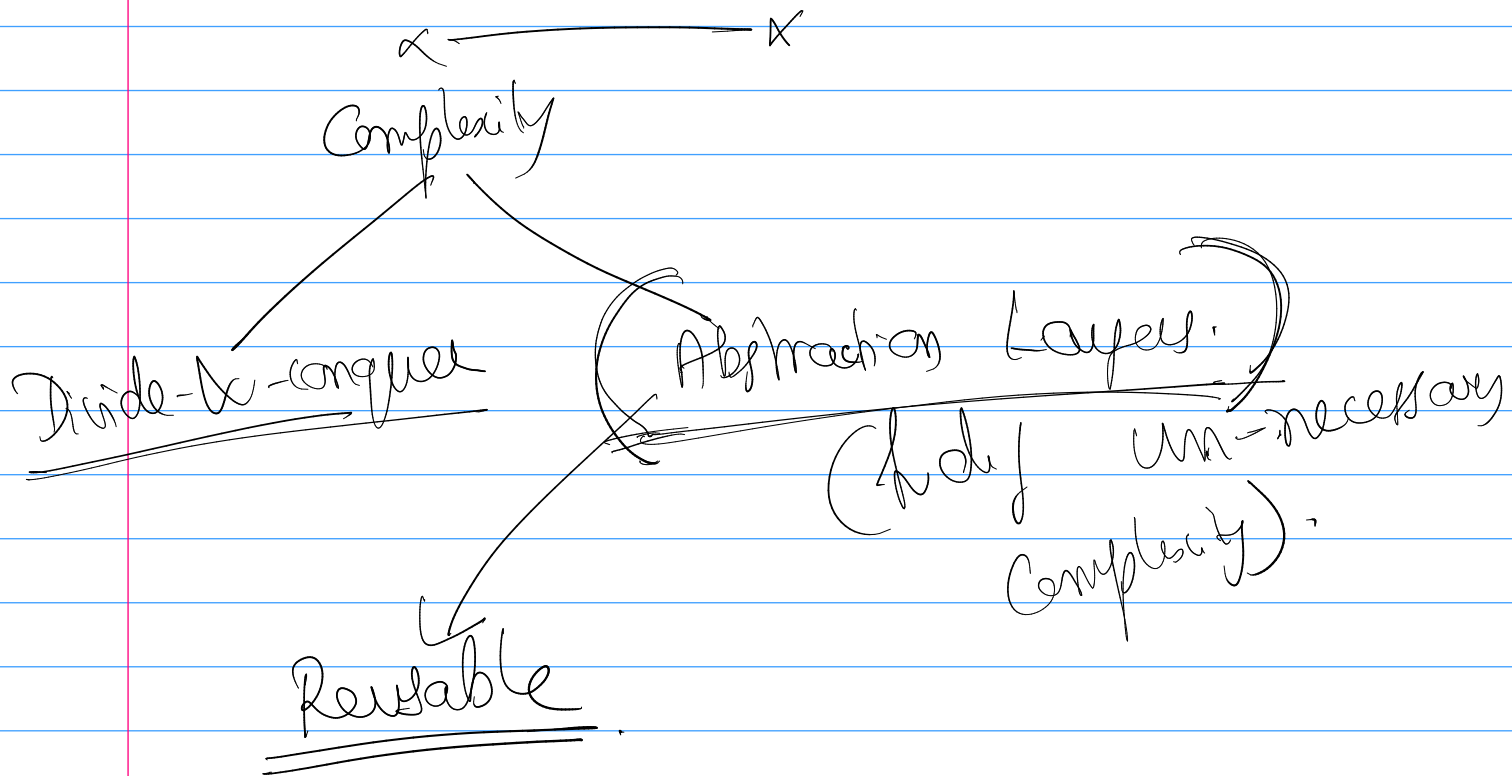
approximate ANY function f^* to
an arbitrary level of accuracy.



$$\forall \epsilon > 0; \exists f \in S$$

$$\therefore \exists \vec{x} : |f(\vec{x}) - f^*(\vec{x})| < \epsilon$$

Depth ^{is} 'Complexity theory' (1990's).
Depth (≥ 2) is 'essential' for
some tasks.



transfer learning